



Weather forecasting in a changing climate: the rise of AI and Machine learning?

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ABSTRACT

Climate change is directly linked to the increase in intensity and frequency of extreme weather events such as heatwaves, heavy rainfall, droughts, floods and hurricanes. In this context, anticipating such extreme events with accurate weather forecasts is of the highest importance in order to provide early warnings to the population and adapt to climate change. Forecasting has long relied on physical models to anticipate atmospheric changes, underpinning decisions that affect millions of lives and livelihoods. In recent years, machine learning has emerged as a powerful new tool, learning from decades of weather data to generate forecasts with remarkable speed and accuracy. Here we show that the integration of machine learning into operational weather prediction, as pioneered by the European Centre for Medium-Range Weather Forecasts, not only matches but regularly surpasses traditional physics-based methods in many metrics. These advances demonstrate that machine learning—far from being a mysterious or threatening ‘black box’—can already be responsibly embedded alongside physics-based methods, boosting predictive skill while fostering new forms of scientific and operational collaboration

1. Introduction

Numerical weather prediction (NWP) has transformed our ability to anticipate atmospheric events, underpinning critical decisions in sectors ranging from disaster management to agriculture (Venuti et al., 2025). Traditionally, forecasts have relied on solving the equations of fluid dynamics and thermodynamics, constrained by observations and implemented on powerful supercomputers (Venuti et al., 2025; Bauer et al., 2015). However, the last five years have witnessed a surge in the application of machine learning (ML) to weather prediction, with models trained on decades of historical data, predominantly based on Copernicus Reanalysis data (Hersbach et al., 2020), now rivalling or surpassing conventional approaches in some settings (Pathak et al., 2022; Bi et al., 2023; Chen et al., 2023; Price et al., 2025).

This shift has been catalysed by the availability of high-quality, openly available reanalysis datasets, the availability of accelerating GPU hardware and advances in deep learning architectures. It has now been demonstrated that, while training an ML model is costly, ML models can generate real-time global forecasts at a fraction of the computational cost of traditional methods (Pathak et al., 2022; Bi et al., 2023; Chen et al., 2023; Price et al., 2025).

This new technique then opens opportunities in operational NWP for which the computer cost and speed of producing forecasts are major factors. It also paves the way for a more equitable playing field, where smaller Meteorological Services, including in the global South, which do not have access to large computer resources, are becoming increasingly empowered. However, in a field which has long relied on physical models, user trust in this new technique cannot be taken as a given, and building trust is an essential element guaranteeing a smooth adoption. The European Centre for Medium-Range Weather Forecasts (ECMWF) has played a key role in this NWP transformation, developing operational ML-based forecasting systems and fostering international collaboration through open-source frameworks (Lang et al., 2024; Raoult et al., 2025; Moldovan et al., 2025; Lang et al., 2025).

2. ML weather forecasting is collaborative

Machine learning weather prediction models are now part of operational production in a growing number of NWP centres, including ECMWF which has, in 2025, made artificial intelligence a core part of its operational forecasting alongside the physics-based forecasts. A key enabler of this progress has been collaboration, in particular through the

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Anemoi framework, an open-source toolkit co-developed with European national meteorological services. Anemoi supports data curation, model training, and operational deployment, fostering a collaborative ecosystem in which advances are rapidly shared and adopted (Raoult et al., 2025). This community-driven approach has led to the development of regional and global ML models, such as Bris (Met Norway) and AICON (Deutscher Wetterdienst, Germany) (Nipen et al., 2025; Prill et al., 2025). The Bris model, originally developed over Norway, has also been tested in different set-ups, in particular over Africa, in collaboration with the Meteorological Service of Malawi. These developments take place in a broader collaboration context, including the ECMWF Machine Learning Pilot Project (ECMWF: 50 years of ECMWF, 2025) and the broader EUMETNET E-AI Artificial Intelligence Programme (E-AI) (EUMETNET, 2025). Beyond ECMWF and collaboration between Met Services, collaboration is also vibrant with research institutes and private companies, for example between the UK Met Office and the Turing Institute (Met Office, 2025) and between the China Meteorological Administration and a broad range of institutions (Chang et al., 2025). In the United States, large collaborative frameworks are also supporting developments in AI/ML (UCAR, 2026; EPIC, 2026).

3. ML weather forecasting works

The performance of ML-based forecasts is now well documented in the peer-reviewed literature. Deterministic ML models have achieved medium-range skill comparable to state-of-the-art NWP, with anomaly correlation coefficients for upper-air fields surpassing or matching those of traditional models out to 10–15 days (Price et al., 2025; Lang et al., 2024). At ECMWF, the Artificial Intelligence Forecasting System (AIFS), implemented in 2025, produces medium-range forecasts using deep neural networks trained on reanalysis and operational data (Lang et al., 2024). AIFS runs alongside the physics-based model of the Integrated Forecasting System (IFS), providing deterministic and ensemble probabilistic forecasts that match or exceed the skill of traditional models in many metrics (Lang et al., 2024; Moldovan et al., 2025; Lang et al., 2025). Examples of performance scores are provided in Fig. 1, for a whole season.

Such improvements are highly significant in a field where advances are hard-won and have huge societal impact. For example, extending the useful forecast range by even a single day can provide earlier warnings for extreme weather, improve disaster preparedness, and enhance decision-making in sectors such as agriculture and energy (Venuti et al., 2025).

Examples where ML forecasts have worked

- Tropical Cyclone Tracks: AIFS surpasses the IFS in predicting the tracks of tropical cyclones (see Fig. 2), capturing both the timing and path of storms with high fidelity (Lang et al., 2024; Moldovan et al., 2025). In some cases, AIFS provided more consistent day-to-day forecasts, reducing the “jumpiness” that can challenge forecasters.
- Sudden Stratospheric Warming (SSW): These events affect surface weather patterns for weeks. During a major SSW event in 2025, the AIFS ensemble predicted the onset and intensity of the event earlier and with greater confidence than the traditional ensemble, with nearly half of ensemble members signalling the event seven days in advance compared to less than 10 % for the conventional system (Moldovan et al., 2025).
- Madden–Julian Oscillation (MJO): ML models have improved the forecast skill for the MJO, extending the useful range at weeks 3–4 in several systems (Chen et al., 2023).
- Heavy Rainfall Events: For day 5 forecasts of heavy rainfall (e.g., exceeding 10 mm/day), ML models have shown slightly better discrimination between rain and dry events, improving early warnings for high-impact precipitation (Moldovan et al., 2025).
- Ability of ML models to forecast out-of-distribution events: during a rare winter snowfall (U.S. Gulf Coast, Jan 2025), as seen in Fig. 3, AIFS captured an extreme, out-of-distribution snowfall/blizzard for this specific region up to 10 days ahead, despite minimal analogues in the training record, demonstrating learned transfer beyond climatology. Similarly, ML models, when learning from strong storm events in only one oceanic basin, have been shown to skilfully forecast strong events in another “unseen” basin (Sun et al., 2025).

Of course, ML forecasts have their limitations – at present they typically have poorer resolution in space and time than physics-based systems, they can be under-dispersive in particular in the case of precipitation, and crucially the range of output products is currently still much more restricted. Nevertheless, the approach of combining information from multiple sources is already well established in meteorological products and services. Accordingly, with ML forecasts having many strengths – and now reliable operational production – they already can and should be used alongside physics-based systems. In this context, it should also be noted that there is currently a clear dependency of the ML forecasts on the physics-based NWP, as most contemporary ML weather models require initial conditions coming from operational analyses of physics-based systems to run inference in real-time.

4. ML weather forecasts: misconceptions and challenges

Despite the evidence, misconceptions persist regarding the role and reliability of AI/ML in weather forecasting. Some commentators have

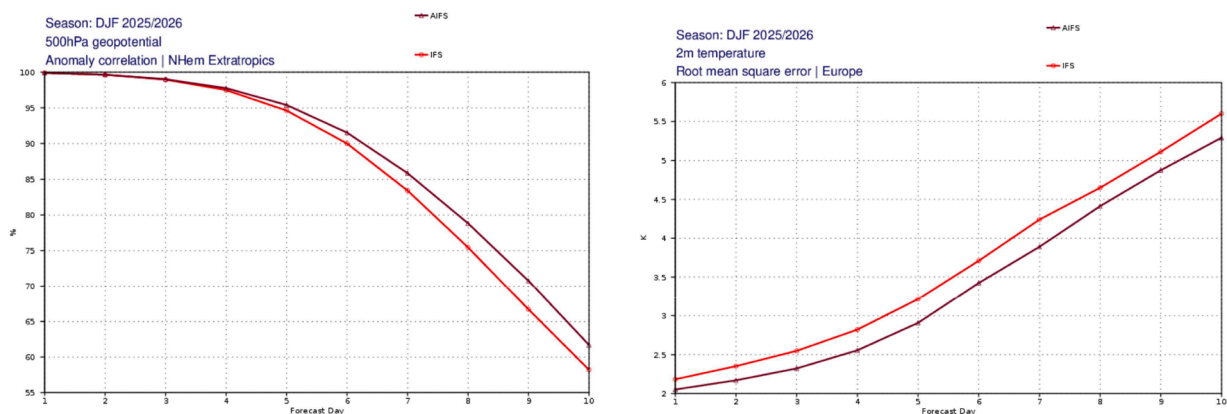


Fig. 1. Comparative verification plots of forecast skill (anomaly correlation and root-mean-square errors) for traditional IFS in red and ML-based AIFS model in purple, illustrating lead time gains and improvements for 500hPa geopotential (first panel) and 2-meter temperature (second panel) averaged over the winter season 2025/2026.

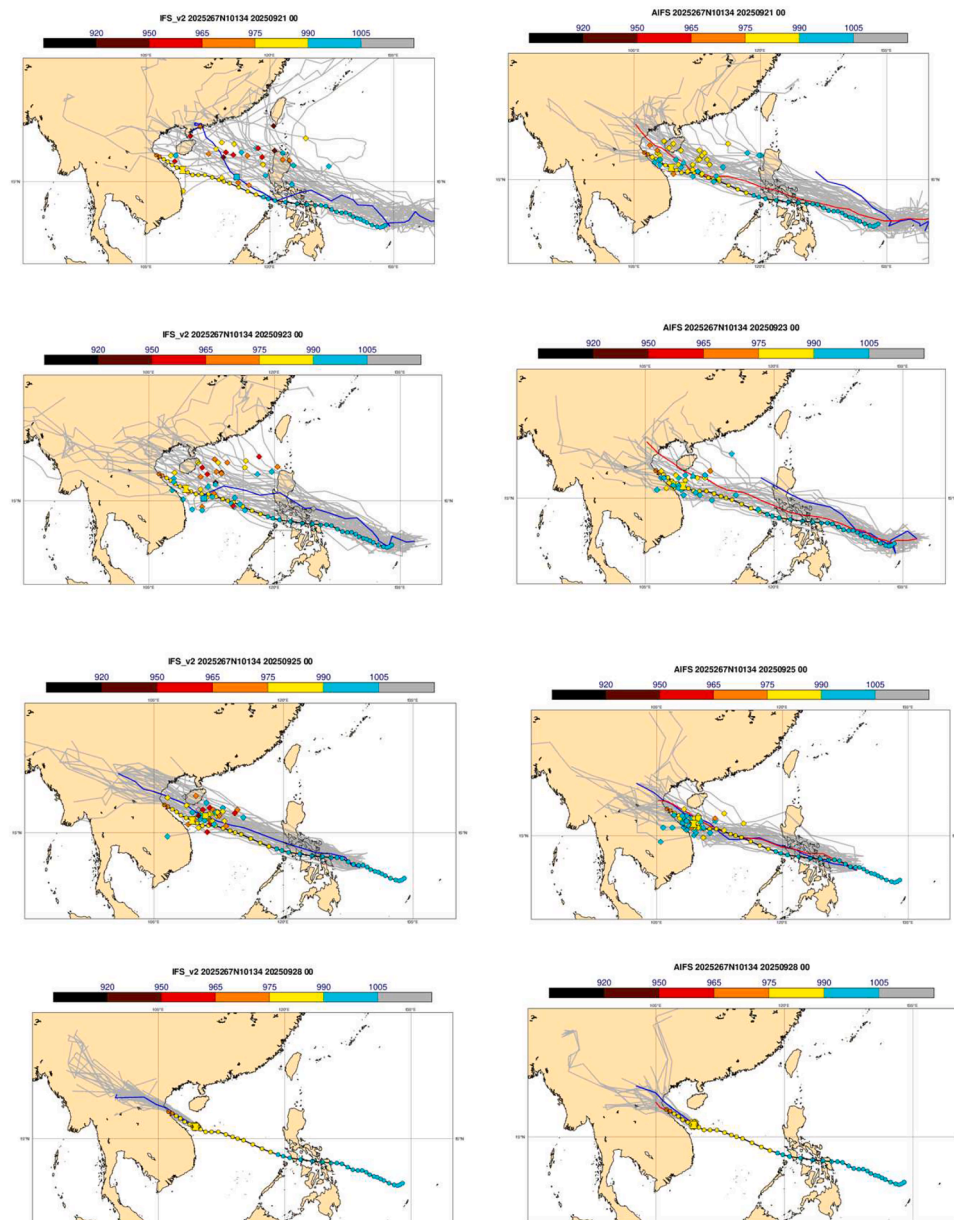


Fig. 2. Forecasts for the tropical cyclone Bualoi which hit the Philippines and Vietnam in September 2025 causing widespread flooding. Ensemble forecasts for the IFS physics-based model (on the left) and the AIFS data-driven model for landfall (on the right) are shown for various forecast ranges ahead of the event. Forecasts are shown for the 28th September at 00UTC, starting respectively from 21st, 23rd, 25th and 28th September, from top to bottom. The analysed best track of the typhoon over its whole life cycle is shown in the solid black line, and the various ensemble forecasts in grey lines. The control forecasts are shown in blue, and the deterministic AIFS forecasts in red. The dots represent the successive positions of the tropical cyclone, coloured following the surface pressure, with a palette indicated above the figure. The yellow hourglass and blue square symbols represent the best track and ensemble control positions, respectively, on 28 September at 00UTC. The diamonds represent the position and intensity of the tropical cyclone for the various ensemble members on 28 September at 00UTC. AIFS is seen to perform better, with the ensemble forecasts more closely representing the track of the typhoon, particularly for the longer ranges.

suggested that ML forecasts are inherently untrustworthy “black boxes” (Reichstein et al., 2019). In practice, the most successful ML systems are those that have been developed transparently, with open sharing of code, weights and data, and with careful validation against established benchmarks (Raoult et al., 2025; Reichstein et al., 2019).

Such is the rapid pace of development that any list of current weaknesses for data-driven methods is in danger of rapidly becoming outdated. Small-scale extreme events are still typically better predicted by physics-based models. But there is room for optimism, as there is scope for significant increase in spatial resolution of data-driven models, powered by datasets such as the upcoming improved version of reanalysis ERA6, Destination Earth digital twin data and observations.

Concerns about the interpretability and physical consistency of ML models are starting to be addressed through the imposition of physical constraints during training, the use of diagnostic tools to interpret outputs (Raoult et al., 2025; Reichstein et al., 2019), and recent evidence demonstrating that ML models are capable of learning physically meaningful atmospheric processes. For instance, comparison with traditional physics-based adjoint analyses has shown that ML models can capture similar physically consistent spatiotemporal linkages between atmospheric variables (Baño-Medina et al., 2025). The low cost and automatic differentiability of ML models also opens up new horizons, such as testing the predictability of a notable heatwave event over North America (Vonich and Hakim, 2024). Such studies contribute to

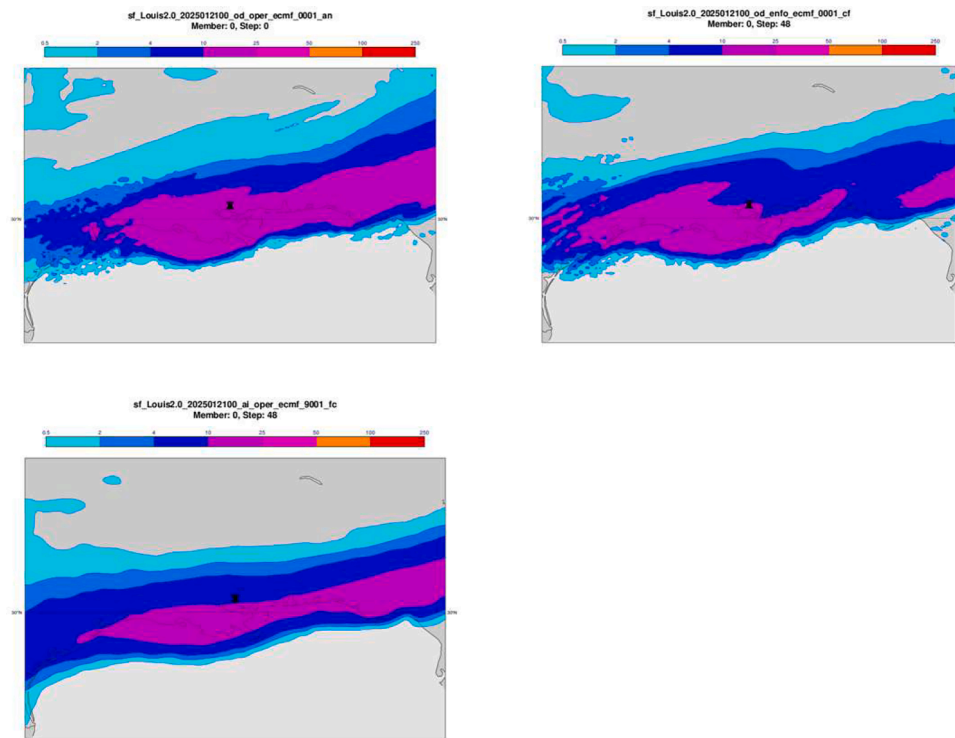


Fig. 3. Precipitation during an exceptional snowfall event over Texas and Louisiana, focussing on 48-hour snowfall from 21 January 00UTC to 23 January 00UTC over the coast. Panels show accumulated precipitation for the 48-hour period, for the verifying analysis (first panel), the 48-hour IFS forecast (second panel) and the 48-hour AIFS forecast (third panel).

the understanding that ML models are not merely black boxes but can exhibit physically interpretable behaviour.

These results, together with the collaborative development of ML models with input from operational forecasters are fostering trust among users and forecasters, who now can routinely incorporate ML guidance alongside traditional model output. ML augments, not replaces, physics-based NWP. Physical models presently offer broader product suites and provide crucial reference and training data; but ML systems are already adding skill and speed (Lang et al., 2024; Moldovan et al., 2025; Lang et al., 2025).

Another misconception is that ML models are only effective because they “learn” from decades of physical model output and thus do not represent a fundamentally new capability or outliers. While it is true that today state-of-the-art ML models rely on high-quality reanalysis data for training, their ability to learn from observation-constrained physical models (via data assimilation) enables data-driven systems to overcome long-standing model biases, such as the slow bias in tropical cyclone tracks or poor wave propagation over orography.

The next generation of data-driven models are beginning to incorporate observations directly, providing opportunities for further improvements and direct forecasting of observed values. More radically, they could also be trained solely based on observations, perhaps one day alleviating the need for reanalysis datasets.

Also, while more work is needed to evaluate the performance of ML models with respect to extremes, it is certainly an over-statement to say that they can only predict what has been seen locally in their training dataset. Moreover, ML models can be rapidly retrained to incorporate new data, adapt to changing climate conditions, and integrate diverse observational sources (Baño-Medina et al., 2025). Beyond these misconceptions, there are certainly some challenges in deploying ML techniques in NWP. First of all, ML models are trained and deployed using an underlying autoregressive rollout, which can have some limitations. For instance, to predict the weather in the medium-range, the model is typically trained over 6-hour periods and optimised for a

72-hour period. Although this seems like a reasonable set-up, it is not suitable to provide high-temporal resolution meteorological fields that will be important for many applications. However, using shorter time-steps leads to greater accumulation of error. Several possible solutions are being explored to alleviate this problem. One tasks the forecasting model to output a series of high-resolution temporal slices in one go. A second approach, downscaling over the temporal dimension (as pioneered by Met Norway in the regional context) is another promising alternative solution.

Another challenge is the need to periodically re-train the model. There might be the need to occasionally re-train the model to tackle climate non-stationarity, but there is also the need to react on a more frequent basis. Operational upgrades to the physics-based systems, which provide the initial conditions for ML models, can change the data distribution away from the training data. Experience at ECMWF has shown that the introduction of a new NWP cycle, if it had not been accompanied by a retuning of the ML model, would have led to a less performing system. This is obviously creating strong dependencies between the two operational systems, with operational consequences.

Possibly the largest, more fundamental, challenges are the full energy footprint and HPC needs of this AI approach and the availability of relevant datasets.

As far as the energy footprint is concerned, it is undeniable that ML training requires significant (GPU) computer resources. These can be obtained from large HPC centres, and in particular Europe is well served with EuroHPC supercomputers managed by the EuroHPC Joint Undertaking. There is also a case to imagine the on demand use of resources from public clouds or the grouping of supercomputer resources between Meteorological Services and European organisations such as ECMWF to cater for the community needs in the future. The governance of federated HPC resources is certainly a topic that will need to be considered. One can imagine that the advent of foundation models, trained for general purposes and then fine-tuned for specific tasks, might make a more optimal use of computing resources, although this needs to be

demonstrated. The European WeatherGenerator project currently under way with many partners across Europe will provide some indications about the effectiveness of this approach.

In any case, although these computer needs are real, one should not underestimate the computational costs of developing, upgrading and maintaining physics-based systems. These are also several orders of magnitude larger than running a single physics-based forecast.

Finally, the challenge of producing, maintaining and making available adequate training datasets, be it fine-scale forecasts, re-analyses or observational datasets, is certainly one that the community should handle, with a clear vision of collective needs, and making all necessary efforts to avoid duplication.

5. Outlook

The integration of machine learning into weather forecasting is reshaping the field, enabling new forms of collaboration, efficiency, and scientific discovery. ECMWF's experience demonstrates that ML weather forecasts can be responsibly embedded within established workflows, delivering tangible improvements in forecast skill. As ML models become standard tools in meteorology, continued emphasis on transparency, collaboration, and rigorous evaluation will be essential to realise their full potential.

Presently, as explained in this article, ML augments, not replaces NWP: the ML prediction step relies on the analysis produced by the physics-based system and the training depends on re-analysis datasets. This obviously creates dependencies and large organisational demands. The exact balance of which system runs routinely in operations and which system is used as a reference and complement is certainly not fixed, and indeed will very likely evolve over time.

Looking ahead, the expansion of ML methods to coupled Earth system modelling, the development of new architectures, and the adoption of continual learning paradigms are likely to further enhance forecast skill and adaptability (Baño-Medina et al., 2025). ECMWF is exploring AI-DOP (McNally et al., 2024; Alexe et al., 2024), an encompassment of data-assimilation and forecasting in a single model that learns to forecast directly from observations (McNally et al., 2024; Alexe et al., 2024). This has shown skilful behaviour, with an array of teams across the globe exploring similar approaches (Sun et al., 2024; Allen et al., 2025; Ni et al., 2025).

Broadly, there appears to be three possible endgames for the AI/ML revolution:

1. Long-term symbiosis of physics-based and ML forecasting systems with complementary strengths.
2. ML forecasting dominates real-time predictions, but physical model data remaining a key ingredient in training these models and physical model developments driving forward ML forecasting skill.
3. ML forecasting dominates real-time predictions with skill improvements being driven by other sources, e.g. observations.

Whilst a future where ML forecasts are the dominant real-time system currently appears plausible, there are significant milestones to achieve this goal: demonstrating superiority for all forecast applications, covering all products and accurately initialising from observations are three key milestones. For the foreseeable future, investment in both physics-based and ML systems appears the optimal path for a robust and accurate weather forecasting ecosystem.

The careful adoption of machine learning in weather prediction stands to benefit both science and society, extending the boundaries of what is possible in forecasting. Data-driven methods are now being successfully deployed in operations for weather forecasting in the short to medium ranges and have started to show potential in the longer seasonal ranges (Kent et al., 2025). Improving weather forecasts through ML methods in the sub-seasonal to seasonal ranges will be particularly important for the adaptation to climate change.

CRedit authorship contribution statement

Florence Rabier: Writing – original draft. **Andrew Brown:** Writing – original draft. **Matthew Chantry:** Writing – original draft. **Florian Pappenberger:** Writing – original draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

References

- Alexe, M., Boucher, E., Lean, P., Pinnington, E., Laloyaux, P., McNally, A., Lang, S., Chantry, M., Burrows, C., Chrut, M., Pinault, F., Villeneuve, E., Bormann, N., Healy, S., 2024. GraphDOP: towards skilful data-driven medium-range weather forecasts learnt and initialised directly from observations. arXiv:2412.15687.
- Allen, A., Markou, S., Tebbutt, W., Requeima, J., Bruinsma, W.P., Andersson, T.R., Turner, R.E., 2025. End-to-end data-driven weather prediction. *Nature* 641 (8065), 1172–1179.
- Baño-Medina, J., Sengupta, A., Doyle, J.D., Reynolds, C.A., Watson-Parris, D., Monache, L.D., 2025. Are AI weather models learning atmospheric physics? A sensitivity analysis of cyclone Xynthia. *NPJ. Clim. Atmos. Sci.* 8 (1), 92.
- Bauer, P., Thorpe, A., Brunet, G., 2015. The quiet revolution of numerical weather prediction. *Nature* 525, 47–55.
- Bi, K., Xie, L., Zhang, H., Chen, X., Gu, X., Tian, Q., 2023. Accurate medium-range global weather forecasting with 3D neural networks. *Nature* 619, 533–538.
- Chang, C.P., Mu, B., Han, W., Ham, Y.G., Zhang, G.J., Damiani, A., Ling, F., Miyoshi, T., 2025. AI weather and climate prediction and applications. In: BAMS meeting summary. <https://doi.org/10.1175/BAMS-D-25-0260.1>.
- Chen, K., Han, T., Gong, J., Bai, L., Ling, F., Luo J.-J., Chen, X., Ma, L., Zhang, T., Su, R., Ci, Y., Li, B., Yang, X., Ouyang, W., 2023. Fengwu: pushing the skilful global medium-range weather forecast beyond 10 days lead. arXiv:2304.02948.
- ECMWF: 50 years of ECMWF, Machine Learning for Numerical Weather Prediction, 2025: <https://www.ecmwf.int/sites/default/files/elibrary/122025/81684-50-years-of-ecmwf.pdf>; <https://www.ecmwf.int/sites/default/files/elibrary/81699-machine-learning-for-numerical-weather-prediction.pdf>. <https://doi.org/10.21957/mpcf-nt20>.
- EPIC, 2026. <https://epic.noaa.gov/nested-eagle-application-release/>.
- EUMETNET, 2025. <https://www.eumetnet.eu/forecasting-and-climate/artificial-intelligence/>.
- Hersbach, H., Bell, B., Berrisford, P., Hirahara, S., Horányi, A., Muñoz-Sabater, J., Nicolas, J., Peubey, C., Radu, R., Schepers, D., Simmons, A., Soci, C., Abdalla, S., Abellan, X., Balsamo, G., Bechtold, P., Biavati, G., Bidlot, J., Bonavita, M., De Chiara, G., Dahlgren, P., Dee, D., Diamantakis, M., Dragani, R., Flemming, J., Forbes, R., Fuentes, M., Geer, A., Haimberger, L., Healy, S., Hogan, R., Hólm, E., Janisková, M., Keeley, S., Laloyaux, P., Lopez, P., Lupu, C., Radnoti, G., de Rosnay, P., Rozum, I., Vamborg, F., Villaume, S., Thépaut, J.-N., 2020. The ERA5 global reanalysis. *Q. J. R. Meteorol. Soc.* 146, 1999–2049. <https://doi.org/10.1002/qj.3803>.
- Kent, C., Scaife, A., Dunstone, N., Smith, D., Hardiman, S., Dunstan, T., Watt-Meyer, O., 2025. Skilful global seasonal predictions from a machine learning weather model trained on reanalysis data. *NPJ. Clim. Atmos. Sci.* 8, 314. <https://doi.org/10.1038/s41612-025-01198-3>.
- Lang, S., Alexe, M., Chantry, M., Dramsch, J., Pinault, F., Raoult, B., Clare, M., Lessig, C., Maier-Gerber, M., Magnusson, L., Ben Bouallègue, Z., Prieto Nemesio, A., Dueben, P., Brown, A., Pappenberger, F., Rabier, F., 2024. AIFS — ECMWF's data-driven forecasting system. arXiv:2406.01465.
- Lang, S., Alexe, M., Clare, M., Roberts, C., Adewoyin, R., Ben Bouallègue, Z., Chantry, M., Dramsch, J., Dueben, P., Hahner, S., Maciel, P., Prieto Nemesio, A., O'Brien, C., Pinault, F., Polster, J., Raoult, B., Tietsche, S., and M. Leutbecher, 2025: AIFS-CRPS: ensemble forecasting using a model trained with a loss function based on the continuous ranked probability score. Preprint. arXiv:2412.15832v1, <https://doi.org/10.48550/arXiv.2412.15832>.
- McNally, A., Lessig, C., Lean, P., Boucher, E., Alexe, M., Pinnington, E., Chantry, M., Lang, S., Burrows, C., Chrut, M., Pinault, F., Villeneuve, E., Bormann, N., Healy, S.,

2024. Data-driven weather forecasts trained and initialised directly from observations. arXiv:2407.15586. <https://arxiv.org/abs/2407.15586>.
- Artificial Intelligence for Numerical Weather Prediction. Combining Met Office and turing expertise to develop a new model - FastNet <https://www.metoffice.gov.uk/research/approach/collaboration/artificial-intelligence-for-numerical-weather-prediction> 2025.
- Moldovan, G., Pinnington, E., Prieto Nemesio, A., Lang, S., Ben Bouallègue, Z., Dramsch, J., Alexe, M., Santa Cruz, M., Hahner, S., Cook, H., Theissen, H., Clare, M., O'Brien, C., Polster, J., Magnusson, L., Mertes, G., Pinault, F., Raoult, B., de Rosnay, P., Forbes, R., and Chantry, M.: AIFS 1.1.0: an update to ECMWF's machine-learned weather forecast model AIFS, EGU sphere [preprint], <https://doi.org/10.5194/egusphere-2025-4716>, 2025.
- Ni, Z., Weyn, J., Zhang, H., Xiang, Y., Bian, J., Jin, W., & Sun, H. (2025). Huracan: a skillful end-to-end data-driven system for ensemble data assimilation and weather prediction. arXiv preprint arXiv:2508.18486.
- Nipen, T., Haugen, H., Ingstad, M., Nordhagen, E., Salihi, A., Tedesco, P., Seierstad, I., Kristiansen, J., Lang, S., Alexe, M., Dramsch, J., Raoult, B., Mertes, G., Chantry, M., 2025. Regional data-driven weather modeling with a global stretched-grid. Artif. Intell. Earth Sys. <https://doi.org/10.1175/AIES-D-25-0001.1>.
- Pathak, J., Subramanian, S., Harrington, P., Raja, S., Chattopadhyay, A., Mardani, M., Kurth, T., Hall, D., Li, Z., Azizzadenesheli, K., Hassanzadeh, P., Kashinath, K., Anandkumar, A. 2022. 'FourCastNet: a global data-driven high-resolution weather model using adaptive fourier neural operators'. <https://doi.org/10.48550/arXiv.arXiv.2202.11214>.
- Price, I., Sanchez-Gonzalez, A., Alet, F., Andersson, T., El-Kadi, A., Masters, D., Ewalds, T., Stott, J., Mohamed, S., Battaglia, P., Lam, R., Willson, M., 2025. Probabilistic weather forecasting with machine learning. Nature 637, 84–90. <https://doi.org/10.1038/s41586-024-08252-9>.
- Prill, F., Jacob, M., DWD AICON Team, 2025. AICON – Introducing ML-based weather forecasting at DWD. ECMWF HPC Workshop. https://events.ecmwf.int/event/464/contributions/5252/attachments/3223/5514/HPC-WS_Prill.pdf.
- Raoult, B., Dramsch, J., Chantry, M., Lang, S., et al., 2025. Anemol: an open-source framework for developing data-driven weather forecasts. In: AMS Annual Meeting abstract, 2025. <https://ams.confex.com/ams/105ANNUAL/meetingapp.cgi/Paper/448707>.
- Reichstein, M., Camps-Valls, G., Stevens, B., Jung, M., Denzler, J., Carvalhais, N., Prabhat, 2019. Deep learning and process understanding for data-driven Earth system science. Nature 566, 195–204.
- Sun, X., Zhong, X., Xu, X., Huang, Y., Li, H., Feng, J., & Qi, Y. (2024). Fuxi weather: an end-to-end machine learning weather data assimilation and forecasting system. arXiv preprint arXiv:2408.05472, 1(5).
- Sun, Y.Q., Hassanzadeh, P., Zand, M., Chattopadhyay, A., Weare, J., Abbot, D.S., 2025. Can AI weather models predict out-of-distribution gray swan tropical cyclones? Proc. Natl. Acad. Sci. USA 122 (21), e2420914122.
- UCAR, 2026. <https://miles.ucar.edu/software/credit/>.
- Venuti, F., Rabier, F., Andersson, E., Modigliani, U., English, S., Kitchen, C., Berrone, M., Pinsault, D., Debrux, J., Jarraud, M., 2025. ECMWF's societal impact through service provision, partnerships and collaborations. J. Eur. Meteorol. Soc. 2, 100013. <https://doi.org/10.1016/j.jemets.2025.100013>.
- Vonich, P.T., Hakim, G.J., 2024. Predictability limit of the 2021 Pacific Northwest heatwave from deep-learning sensitivity analysis. Geophys. Res. Lett. 51 (19), e2024GL110651.